

DATA STRUCTURE ALL PYQ AKTU

✓ UNIT 1: Introduction to Data Structures, Arrays, and Complexity

Define Data Structure. List linear and non-linear structures with applications. (2017–18)

Define Abstract Data Type (ADT) and its importance. (2024–25)

What is Time-Space Tradeoff? Explain with example. (2018–19, 2024–25)

Define and compare Big O, Omega, and Theta notations. (2024–25, 2019–20)

Derive Big O for Linear Search. (2018–19)

Differentiate between arrays and linked lists. (2021–22)

Represent Sparse Matrix in memory. (2019–20, 2024–25)

Compute address of $\text{arr}[i][j]$ in row/column major. (2023–24, 2018–19, 2022–23)

Difference between overflow and underflow in linked list. (2018–19)

Write algorithm to concatenate two linked lists. (2023–24)

✓ UNIT 2: Stack, Queue and Recursion

Explain Stack with applications. (2021–22, 2024–25)

Stack operations using array/linked list (push, pop). (2019–20, 2024–25)

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Convert infix to postfix: $A+B*C-D/F$. (2023–24, 2018–19, 2022–23)

Evaluate expression using stack: $3*(5-3)$. (2023–24)

Check for balanced parentheses using stack (C program). (2024–25)

Queue using array. Full/Empty conditions for circular queue. (2021–22, 2018–19, 2022–23)

Represent dequeue using linked list (insert/delete at both ends). (2023–24)

Explain priority queue and its application. (2024–25, 2020–21)

Discuss recursion with example. Tower of Hanoi problem. (2018–19, 2021–22)

Tail recursion and its advantages. (2020–21, 2021–22, 2022–23)

✓ UNIT 3: Linked Lists

Operations on singly/doubly linked list (insert/delete/traverse). (2019–20, 2024–25)

Write C code to delete specific element in single linked list. (2018–19)

Advantages of doubly linked list over singly linked list. (2021–22, 2022–23)

Represent polynomial using linked list + add 2 polynomials. (2024–25, 2021–22, 2022–23)

DATA STRUCTURE ALL PYQ AKTU

Logical representation of stack using linked list. (2024–25)

Concatenation of two linked lists (pseudo code). (2023–24)

✓ UNIT 4: Trees

Define Binary Tree, Full, Complete, Extended Binary Tree. (2020–21)

Binary Tree Traversals: Preorder, Inorder, Postorder. (2017–18, 2023–24, 2022–23)

Construct binary tree from preorder & inorder sequences. (2018–19, 2024–25)

Expression Tree construction from infix/postfix. (2018–19, 2022–23)

Construct AVL Tree from given elements. (2018–19, 2024–25, 2022–23)

Threaded Binary Tree & its advantages. (2024–25, 2020–21)

Difference between internal and external path length. (2023–24)

Represent B-tree of given order; insert & delete elements. (2018–19, 2024–25, 2022–23)

✓ UNIT 5: Graphs and Hashing

Represent Graph in memory (Adjacency matrix/list). (2018–19, 2024–25)

BFS and DFS with examples. (2018–19, 2024–25, 2022–23)

Connected components algorithm. (2023–24, 2022–23)

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Shortest path using Dijkstra's algorithm. (2018–19, 2023–24, 2022–23)

MST using Kruskal's / Prim's algorithm. (2018–19, 2024–25, 2022–23)

Hashing techniques & collision resolution. (2020–21, 2023–24, 2022–23)

Linear vs Quadratic Probing (with disadvantages). (2023–24)

Degree, indegree/outdegree in graph (code). (2023–24)

Warshall / Floyd-Warshall algorithm with application. (2018–19, 2021–22, 2022–23)

Huffman Coding: significance and algorithm. (2018–19, 2023–24, 2022–23)